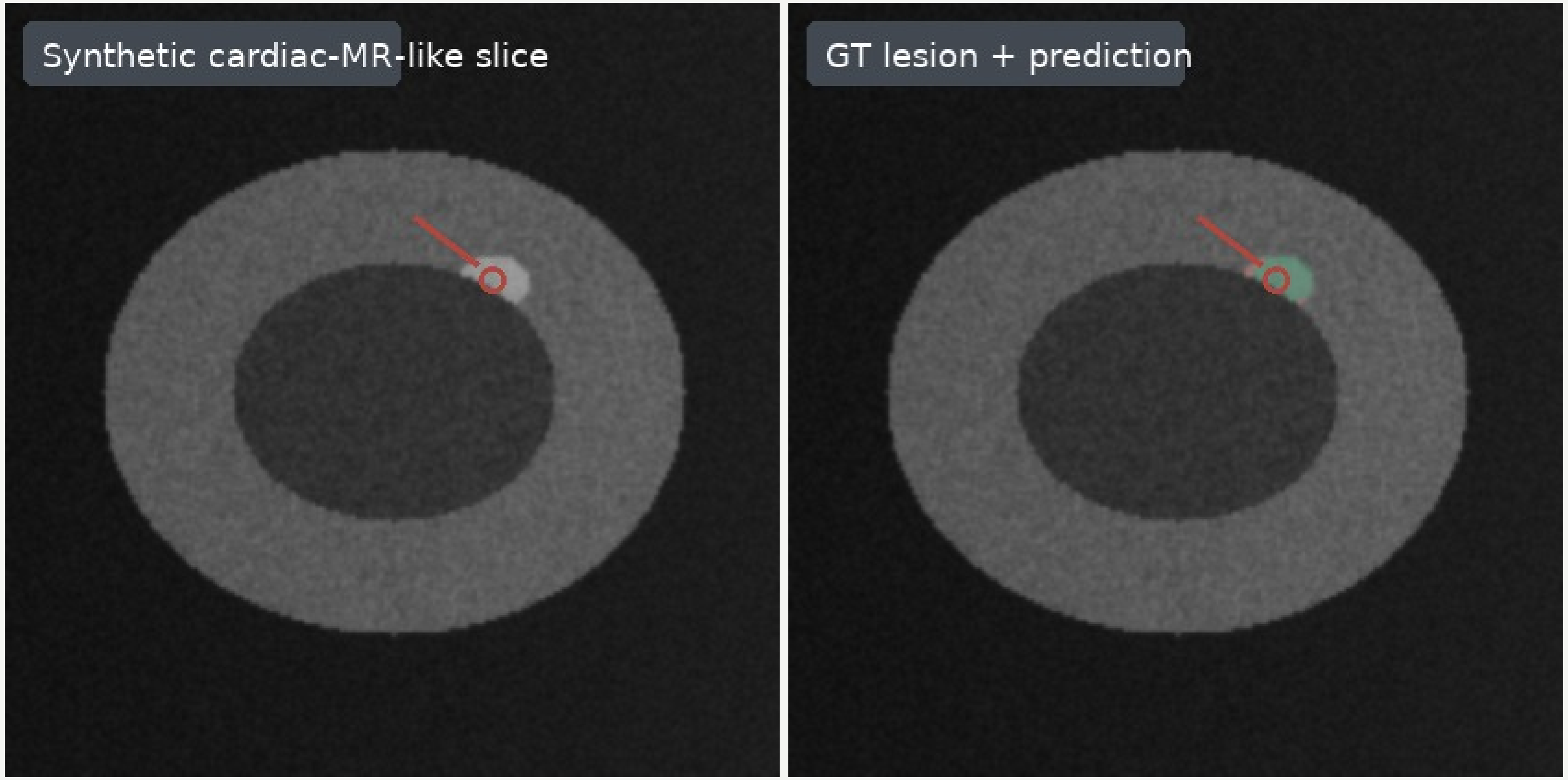
A small lesion is the unit of failure

Synthetic cardiac-MR-like segmentation task; not clinical validation



GT
Prediction / TP
FP
FN

Evaluation unit: lesion detection + mask overlap

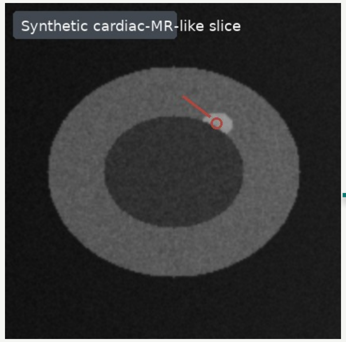
Dice summarizes mask overlap, but a missed small lesion is counted at the lesion level.

Endpoints: Dice overlap, lesion recall, false-positive burden

Synthetic cardiac-MR-like phantoms; not clinical validation.

The stress test follows one image-to-endpoint path

Center appearance shift propagates through prediction before aggregation



synthetic image + GT lesion → **deterministic prediction** → **case metrics Dice, recall, FP** → **center summary + lesion strata**

3 center conditions

contrast, noise, bias

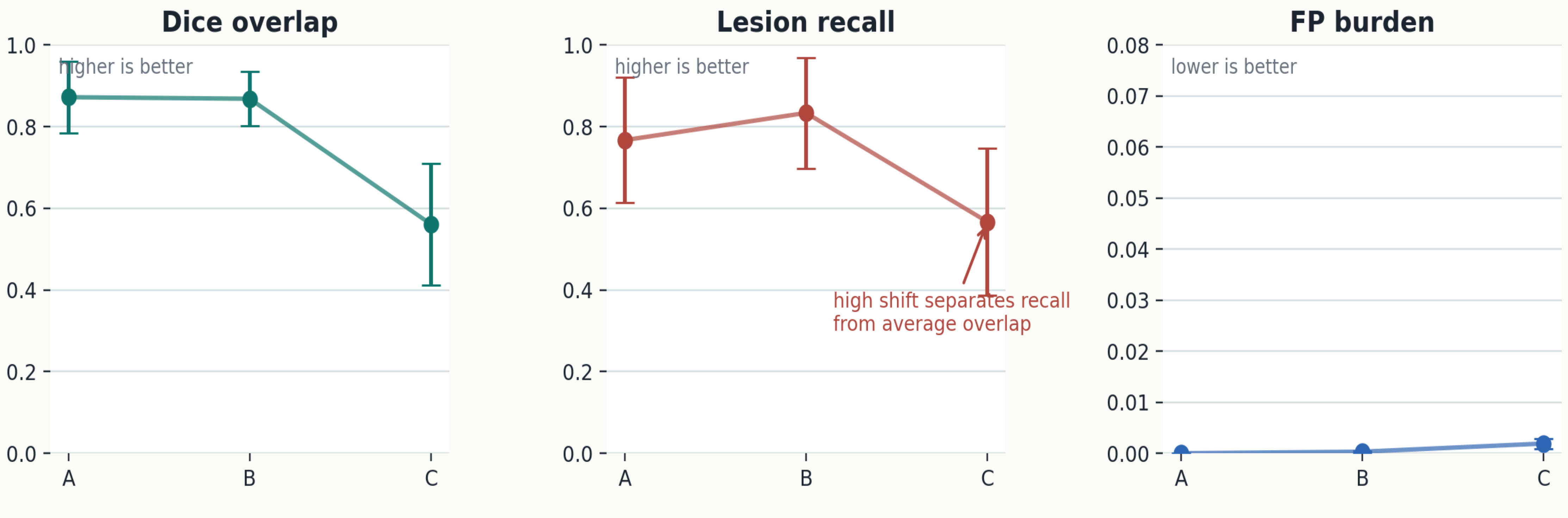
Case-level metrics stay attached to the same image, GT and prediction before any center-level average.

The stress test asks whether center-level average Dice hides lesion-level failures under appearance shift.

Aggregation is descriptive: no clinical validation claim is made from these phantoms.

Average Dice can hide a lesion-level failure

Endpoint disagreement appears under the high-shift center



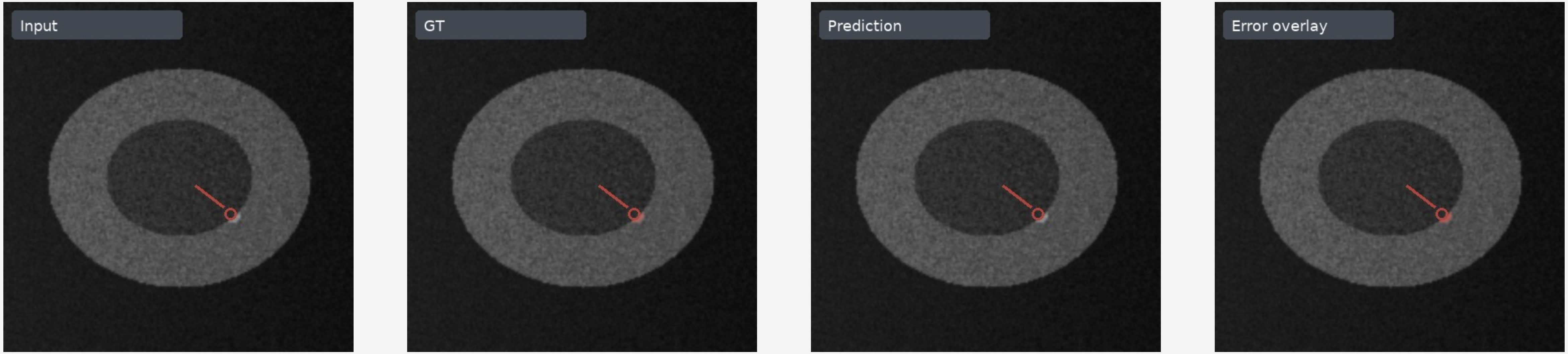
Mean across deterministic synthetic cases; vertical intervals show approximately 95% cross-case uncertainty.

Center C average Dice = 0.56, but small-lesion recall drops to 0.00.

FP burden is lower-is-better; Dice and recall are higher-is-better.

The failure is visible in the same-case overlay

A missed small lesion drives the negative endpoint



Input | GT lesion | Prediction | TP / FP / FN

GT   Prediction / TP   FP   FN

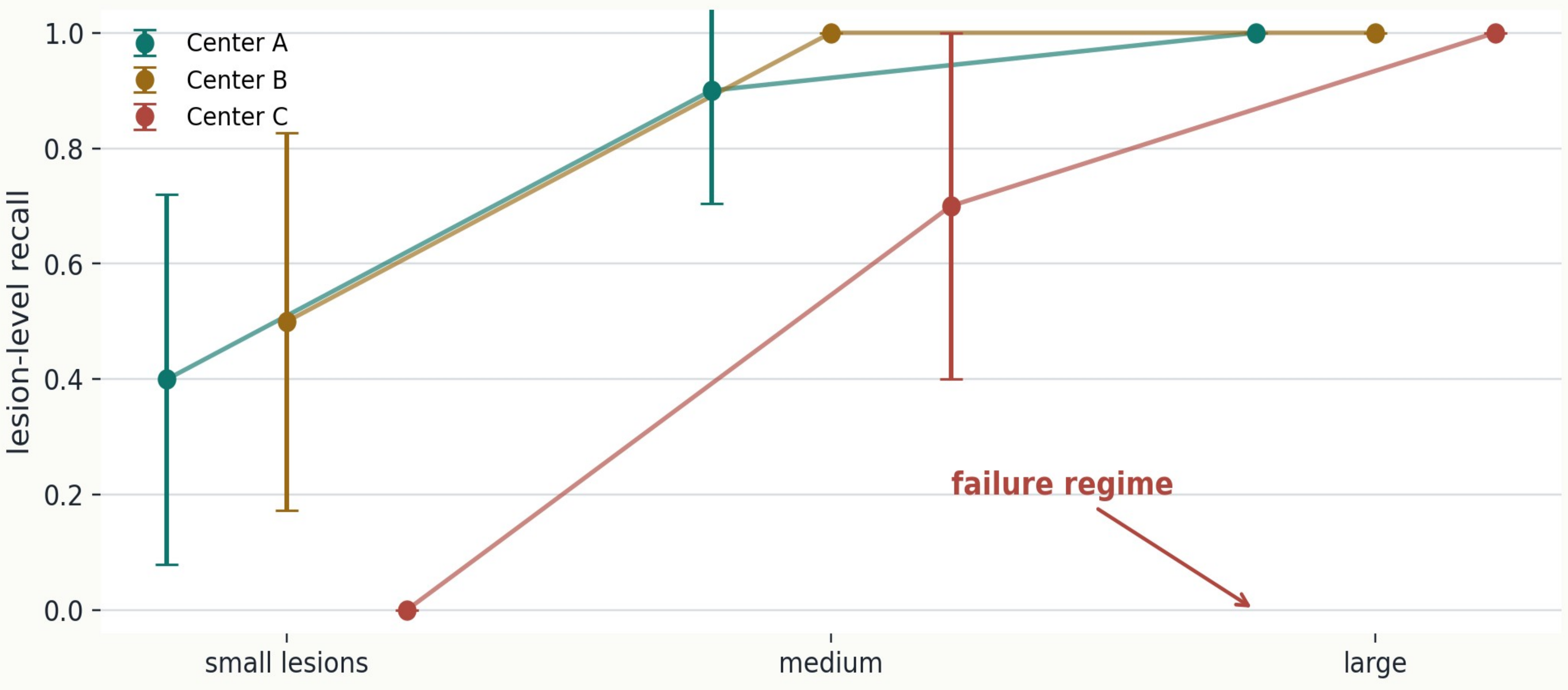
Case C3-03 (Center C, small lesion): Dice 0.00; lesion recall 0; FN pixels 20.

The red FN region shows why an average overlap metric is not enough for this stress condition.

All four panels are the same synthetic slice geometry.

Validate the small-lesion shift explicitly next

Completed synthetic evidence motivates a planned held-out-center experiment



Completed evidence: deterministic synthetic phantoms only; next validation is planned, not completed.

Completed evidence

Small lesions in the high-shift center are the supported failure regime in this deterministic phantom benchmark.

Planned validation

Hold out center C, stratify by lesion size, and sweep prediction threshold/calibration before comparing methods.

Synthetic cardiac-MR-like phantoms; not a patient validation endpoint.